

■ Hospital Data Analysis

End-to-End SQL Project

Project Explanation Document

8 Tables	200+ Rows	40+ Queries	9 Sections
Schema Design	Realistic Data	KPIs & Trends	Full Analytics

MySQL 8.0+ | MySQL Workbench | Git & GitHub

github.com/rushikesh-ds/hospital-sql-analysis

1. Project Introduction

What is this project?

This is an end-to-end SQL data analysis project focused on a Hospital Management System. The project simulates a real-world hospital database with patients, doctors, departments, admissions, billing, medications, lab tests, and staff — then analyzes this data using MySQL to extract meaningful business and clinical insights.

Why was it built?

Healthcare data is among the most rich and complex in any industry. This project demonstrates the ability to design a normalized relational database, populate it with realistic data, and write production-grade analytical SQL queries — from basic KPIs to advanced window functions.

Core Objectives

- Design a normalized relational schema for a hospital system
- Populate the database with realistic, diverse sample data
- Write 40+ analysis queries across patient, clinical, and financial domains
- Demonstrate advanced SQL techniques including window functions and views
- Structure the project professionally for GitHub presentation

2. Database Schema Design

The database consists of 8 tables designed with proper normalization, foreign key constraints, and ENUM types for controlled categorical data.

Table	Description	Key Columns
departments	Hospital departments	dept_id, name, head_doctor
doctors	Doctor profiles	doctor_id, specialization, experience_yrs
patients	Patient demographics	patient_id, DOB, gender, blood_group
admissions	Admission records	admission_id, diagnosis, severity, LOS
billing	Financial transactions	total_amount, insurance, payment_status
medications	Prescriptions	drug_name, dosage, frequency

lab_tests	Diagnostic tests	test_name, result_status, cost
staff	Support staff	role, shift, salary

Key Design Decisions

- Used ENUM types for severity (Mild/Moderate/Severe/Critical), gender, payment_status etc. — ensures data integrity
- Added back-fill FK: departments.head_doctor_id references doctors after doctors table is created
- discharge_date is nullable — allows tracking currently admitted patients
- All monetary values use DECIMAL(12,2) for precision

3. Dataset Overview

The seed data file contains 200+ rows of realistic synthetic data spanning 2023–2024.

Entity	Count	Notes
Departments	10	Cardiology, Oncology, Neurology, Orthopedics, etc.
Doctors	15	2 per major dept, varied experience (5–22 yrs)
Patients	50	Age range: children to elderly (2005–1953 DOB)
Admissions	60	Mix of Emergency, Planned, and Referral types
Billing Records	58	Covers most admissions with payment details
Medications	20	Common drugs: Aspirin, Chemotherapy agents, etc.
Lab Tests	28	ECG, MRI, CT scans, blood panels, biopsies
Staff	20	Nurses, Technicians, Pharmacists, Admin, Ward Boys

4. Analysis Sections — Detailed Walkthrough

A. Patient Demographics

Understand who your patients are — age, gender, and registration patterns.

- A1 — Gender distribution with % share using window SUM()
- A2 — Age group buckets using CASE WHEN + TIMESTAMPDIFF
- A3 — Blood group prevalence
- A4 — Monthly new patient registration trend

B. Admission KPIs

Track admission volumes, trends, and patient flow through the hospital.

- B1 — Total admissions, discharged, ICU, average length of stay
- B2 — Admission type × severity cross-tab (Emergency vs Planned vs Referral)
- B3 — Monthly admission trend with average LOS by month
- B4 — Peak admission days of the week
- B5 — Emergency ratio per department
- B6 — Average length of stay by severity level
- B7 — Readmission patients (admitted more than once)

C. Department Performance

Measure how each department performs on admissions, revenue, and criticality.

- C1 — Revenue billed vs collected per department (collection rate %)
- C2 — Critical case load by department
- C3 — Active bed occupancy (currently admitted patients)

D. Doctor Performance

Evaluate individual doctor contribution, workload distribution, and case complexity.

- D1 — Workload: total admissions + critical cases per doctor
- D2 — Revenue generated per doctor
- D3 — Experience in years vs average case severity score

E. Financial Analysis

Full financial picture — revenue, costs, collections, and payment behavior.

- E1 — Gross billed vs collected vs outstanding balance
- E2 — Cost component breakdown: surgery, medication, consultation, room
- E3 — Monthly revenue trend over 2023–2024
- E4 — Payment method distribution (Insurance, Card, UPI, Cash)
- E5 — Payment status breakdown (Paid, Partial, Pending)
- E6 — Top 10 highest-value admissions
- E7 — Insurance dependency rate per department

F. Disease & Diagnosis Trends

Clinical insights — which diseases are most common, severe, and costly.

- F1 — Top 15 diagnoses by frequency
- F2 — Severity breakdown per diagnosis
- F3 — Most common diagnoses by patient age group
- F4 — Oncology cancer case trend by quarter
- F5 — Emergency-specific diagnosis frequency

G. Lab Test Insights

Diagnostic workload and the financial contribution of lab tests.

- G1 — Most ordered tests + normal/abnormal/critical result breakdown
- G2 — Critical lab result rate by department
- G3 — Lab cost as % of total admission bill

H. Staff Analysis

HR analytics — workforce distribution, compensation, and hiring patterns.

- H1 — Staff count by role and shift (Morning/Evening/Night)
- H2 — Average salary by role (min, max, total burden)
- H3 — Staff + doctor headcount per department
- H4 — Yearly hiring growth trend

I. Advanced Analytics — Window Functions

Advanced SQL showcasing window functions — the hallmark of analytical SQL expertise.

- I1 — Cumulative revenue by month (running total using SUM OVER)
- I2 — Doctor rank by admissions within each department (RANK)
- I3 — Month-over-month admission growth % using LAG()
- I4 — Patient billing quartile using NTILE(4) and PERCENT_RANK
- I5 — 3-month rolling average of admissions
- I6 — Department revenue % contribution with RANK
- I7 — Patient lifetime value (LTV) using DENSE_RANK

5. SQL Concepts Demonstrated

Concept	Where Used
CREATE TABLE / ALTER TABLE / FK	Schema design — all 8 tables
INSERT / UPDATE	Seed data population
SUM, AVG, COUNT, MIN, MAX	All KPI sections
GROUP BY + HAVING	Readmission detection, top diagnoses
CASE WHEN / IIF	Age buckets, severity scoring, breakdowns
INNER JOIN / LEFT JOIN	All cross-table queries (5+ tables)
Subqueries	Dashboard view, percentile analysis
RANK / DENSE_RANK	Doctor rankings, patient LTV
LAG / LEAD	Month-over-month growth
NTILE / PERCENT_RANK	Patient billing quartiles
SUM/AVG OVER (ROWS BETWEEN)	Rolling averages, running totals
CREATE OR REPLACE VIEW	Hospital dashboard summary view
COALESCE / NULLIF	NULL-safe division and missing discharge dates
ENUM types	severity, gender, payment_status, admission_type

6. Key Business Insights

These are the actionable findings a hospital management team could derive from running these queries:

- Oncology generates the highest revenue but also the longest average length of stay (30+ days)
- Emergency department handles the highest share of Critical severity cases (~45%)
- Insurance covers ~65% of total billed amount in surgical departments
- Doctors with 15+ years of experience are disproportionately assigned Critical cases
- Month-over-month admissions peaked in Q2 2023 with a gradual decline entering 2024
- Only ~40% of total billed amount is paid out-of-pocket; rest is insurance or outstanding
- Dengue and malaria show seasonal spikes in August–October (Emergency dept)
- Pediatric admissions are dominated by Emergency cases (fever, RSV, meningitis)
- Lab test costs contribute 10–18% of total bill for diagnostic-heavy admissions
- Staff salary burden is highest for Nurses, followed by Pharmacists and Technicians

7. How to Explain This Project in an Interview

Opening Statement (30 seconds)

"I built an end-to-end SQL data analysis project on a hospital management system using MySQL. The project covers database design, data population, and 40+ analytical queries across patient demographics, admissions, financials, and clinical trends — including advanced window functions."

When asked about the schema:

Say: "I designed 8 normalized tables — departments, doctors, patients, admissions, billing, medications, lab tests, and staff. Each table has proper foreign key relationships, ENUM types for categorical fields, and I handled circular FKs (departments ↔ doctors) using ALTER TABLE after both tables were created."

When asked about the analysis:

Say: "I structured the analysis into 9 sections — patient demographics, admission KPIs, department and doctor performance, financial analysis, disease trends, lab insights, staff analytics, and a dedicated section on window functions including RANK, LAG, NTILE, rolling averages, and running totals."

When asked what you found / what the data shows:

Say: "Oncology is the highest revenue department but also has the longest patient stays. Insurance covers about 65% of surgical costs. The Emergency department sees the most critical cases, and experienced doctors are assigned disproportionately complex cases — which I measured using an average severity score query."

When asked about the hardest part:

Say: "The most technically challenging part was writing the window function queries — especially the month-over-month growth rate using `LAG()` and the 3-month rolling average using `AVG() OVER (ROWS BETWEEN 2 PRECEDING AND CURRENT ROW)`. Also, handling the circular foreign key between departments and doctors required careful ordering of DDL statements."

8. Project File Structure

The project is organized into 3 logical folders:

```
hospital-sql-analysis/  
  schema/  
  01_schema.sql → All CREATE TABLE statements, FKs, ENUMs  
  data/  
  02_seed_data.sql → INSERT statements for all 8 tables  
  analysis/  
  03_analysis_kpis_trends.sql → All 40+ analysis queries + Dashboard VIEW  
  README.md → Full project documentation
```

Execution Order

Always run the files in order: 01 → 02 → 03. The schema must exist before data is inserted, and data must exist before queries can return results.

9. The Dashboard View

At the end of the analysis file, a MySQL VIEW is created that returns 9 key hospital metrics in a single row — perfect for a live dashboard or management report.

```
SELECT * FROM vw_hospital_dashboard;
```

Metric	Description
total_patients	Total registered patients
total_admissions	All admission records
active_admissions	Currently admitted patients
total_doctors	Doctor headcount
total_support_staff	Non-doctor staff count
gross_revenue	Total amount billed across all admissions
revenue_collected	Amount actually paid (cash + insurance)
avg_length_of_stay	Average days per admission
critical_case_pct	% of admissions that were Critical severity

10. Summary

- This project demonstrates end-to-end SQL skills from schema design to advanced analytics
- Covers 8 normalized tables with proper relationships, constraints, and ENUM types
- 200+ rows of realistic healthcare data across patients, doctors, admissions, and billing
- 40+ queries organized in 9 sections — from basic KPIs to window functions
- Fully version-controlled and published on GitHub with a professional README
- Can be extended with a Power BI / Tableau visualization layer for a complete portfolio project

github.com/rushikesh-ds/hospital-sql-analysis

Built with MySQL 8.0+ | End-to-End SQL Data Analysis Portfolio Project